3. Analyse and apply the appropriate automated manufacturing support system.

Recommended Books

1. Koren, Y., Computer Control of Manufacturing systems, McGraw Hill (2009).
2. Suh Suk-Hwan, Kang Seong-Kyoon, Chung Dae-Hyuk, Stroud Ian., Theory and Design of CNC Systems, 2008, Springer-Verlag London Limited
3. Smith Peter, CNC programming handbook, 2^{nd} edition, 2003, Industrial Press Inc.
4. Groover, M. P. and Zimmers, E. W., CAD/CAM:Computer Aided Design & Manufacturing, 2006, Pearson Education India
5. Hood-Daniel P., and Kelly J.F., Build Your Own CNC Machine, 2009, Springer-Verlag New York
6. Manuals of CAD/CAM Software Package on CAM Module and CNC Machines.

Evaluation Scheme

S. No.	**Evaluation Elements**	**Weightage (%)**
1.	MST	25
2.	EST	40
3.	Sessionals (Research Assignments/Projects/Tutorials/Quizes)	35

UME748 : Advanced Internal Combustion Engines

L	T	P	Cr
3	0	0	3.0

Course Objective: The students will learn to classify different types of internal combustion engines and their applications. Students will be exposed to fuel air cycles, combustion charts, two stroke engines. The students will study fuel supply systems in SI and CI engines, dual fuel and multi fuel engines, alternative fuels. Detailed study will be done on recent trends in IC engines, emission control strategies.

Syllabus

Introduction: Thermodynamic properties of fuel-air mixture before and after combustion, deviations of actual cycle from Ideal conditions, Analysis using combustion charts.

S.I. Engines: Design of carburation system, MPFI, combustion, ignition systems, Combustion chambers in S.I. engines.

C.I. Engines: Fuel injection, combustion, swirl & inlet ports design, Combustion: DI models, Supercharging, turbocharging & matching of turbocharging.

Engine Lubrication: Friction and lubrication, Performance, Two stroke engine: scavenging, standards.

Recent Trends in I.C. Engines: Dual-fuel engines, multi-fuel engines, stratified charge engine, Stirling engine, variable compression ratio engine.

Engine Emissions & Control Air pollution due to IC engines, engine emissions, exhaust gas recirculation, modern control strategies, Engine emissions standards and norms.

Research assignment: Preparation of Diesel emulsion with nanoparticles, biofuel and check for thermal, physical, chemical properties of fuel and emission characteristics at various loads. Examples of spark ignition and compression ignition engines and new technologies involve in fuel supply systems. Waste heat recovery in IC engines. Design of simple carburetor

Course Learning Objectives (CLO)

The students will be able to:

1. Analyze the engine thermodynamic characteristics using fuel air cycles and combustion charts.
2. Evaluate and analyze the parameters in the engine for issues of power generation, emissions and environmental impact, fuel economy.
3. Analyze the effects of fuel composition on engine operation and mechanical limitations for ideal performance.
4. Analyze the air induction and fuel supply processes for both SI and CI engines.
5. Analyze the effect of spark timing, valve timing and lift, cylinder dimensions, compression ratio, combustion chamber design shape.

Text Books

1. Pulkrabek, W. W., Engineering Fundamentals of the Internal Combustion Engines, Pearson Education, New Delhi (2007).
2. Heisler, H., Advance Engine Technology, Butter Worth Hienemann, USA (2000)
3. Stone, R., Introduction to Internal Combustion Engines, Pearson Education, New Delhi(1999).